

MAT1575, Classwork2, Fall2026

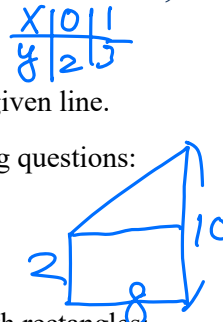
1.1 Approximating Areas (p.6-23)

1. Example: Approximating Area of a Trapezoid under a given line.

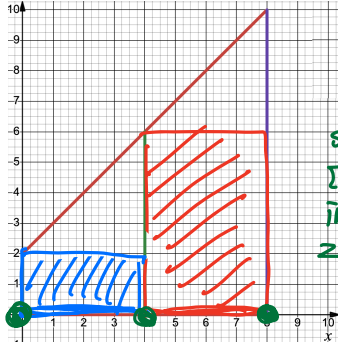
Given a function $f(x) = x + 2$. Please finish the following questions:

(a) Find the area between $y = 0$ and $f(x)$ for $0 < x < 8$.

Area $A = \frac{(2+10) \cdot 8}{2} = 48$

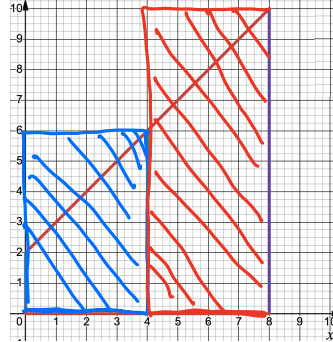


(b) Estimate the area in (a) by dividing it into 2 equal-width rectangles:



Left-endpoint approximation

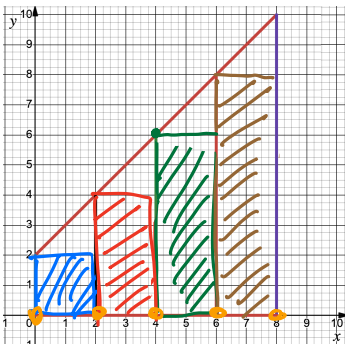
$L_2 = 2 \times 4 + 6 \times 4$
 $\text{splitting} = 8 + 24$
 $[0,8] = 32$
 into 2 parts



Right-endpoint approximation

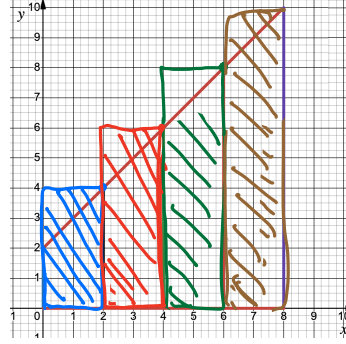
$R_2 = 6 \times 4 + 10 \times 4$
 $= 24 + 40$
 $= 64$

(c) Estimate the area in (a) by dividing it into 4 equal-width rectangles:



Left-endpoint approximation

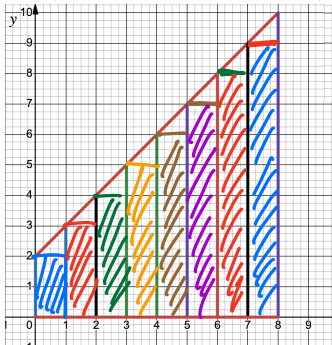
$L_4 = 2 \times 2 + 4 \times 2 + 6 \times 2 + 8 \times 2$
 $= 4 + 8 + 12 + 16$
 $= 40$



Right-endpoint approximation

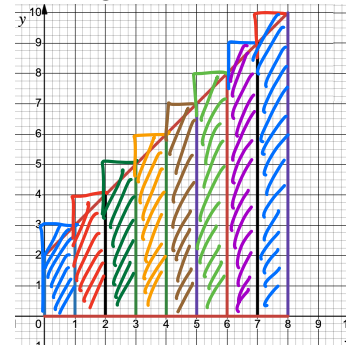
$R_4 = 4 \times 2 + 6 \times 2 + 8 \times 2 + 10 \times 2$
 $= 56$

(d) Estimate the area in (a) by dividing it into 8 equal-width rectangles:



Left-endpoint approximation

$L_8 = 2 \cdot 1 + 3 \cdot 1 + 4 \cdot 1 + 5 \cdot 1 + 6 \cdot 1 + 7 \cdot 1 + 8 \cdot 1 + 9 \cdot 1$
 $= 44$



Right-endpoint approximation

$R_8 = 3 \cdot 1 + 4 \cdot 1 + 5 \cdot 1 + 6 \cdot 1 + 7 \cdot 1 + 8 \cdot 1 + 9 \cdot 1 + 10 \cdot 1$
 $= 52$

(e) What is the relationship between $A, L_2, R_2, L_4, R_4, L_8$, and R_8 ?

observation, if we have L_n (or R_n), what if $n \rightarrow \infty$

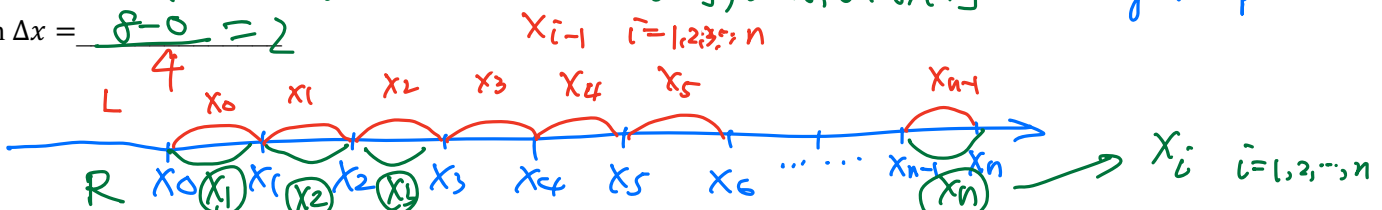
2. Definition: Partition

A set of points $P = \{x_i\}$ for $i = 1, 2, \dots, n$ with $a = x_0 < x_1 < x_2 < \dots < x_n = b$, which divides the interval $[a, b]$ into subintervals of the form $[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n]$ is called a partition of $[a, b]$.

If the subintervals all have the same width, the set of points forms a regular partition of the interval.

Example in 1(b): $P = \{0, 4, 8\}$ it divides $[0, 8]$ into $[0, 4]$ and $[4, 8]$ which is a regular partition and width $\Delta x = \frac{8-0}{2} = 4$

Example in 1(c): $P = \{0, 2, 4, 6, 8\}$ it divides $[0, 8]$ into $[0, 2], [2, 4], [4, 6], [6, 8]$ which is a regular partition and width $\Delta x = \frac{8-0}{4} = 2$



$P = \{x_0, x_1, x_2, x_3, \dots, x_n\}$, regular partition $\Delta x = \frac{x_n - x_0}{n}$

3. Left-endpoint approximation L_n and Right-endpoint approximation R_n

On each subinterval $[x_{i-1}, x_i]$ (for $i = 1, 2, \dots, n$), we construct a rectangle with width Δx and height h to calculate the

area of this rectangle is $h \cdot \Delta x$. Adding the areas of all these rectangles, we get an approximation of the area

(a) Left-endpoint approximation L_n

$h = f(x_{i-1})$ which is the function value at the left endpoint of the subinterval and

$$L_n = (f(x_0) + f(x_1) + f(x_2) + \dots + f(x_{n-1})) \cdot \Delta x$$

$$= \sum_{i=1}^n f(x_{i-1}) \cdot \Delta x$$

sum

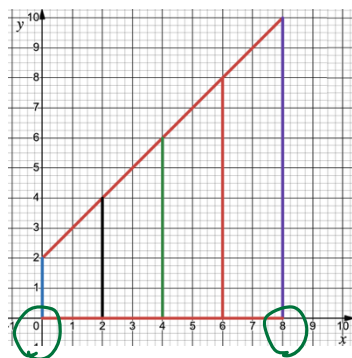
(b) Right-endpoint approximation R_n

$h = f(x_i)$ which is the function value at the right endpoint of the subinterval and

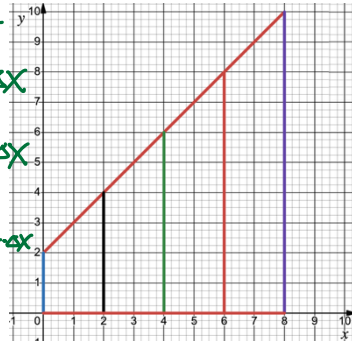
$$R_n = f(x_1) \cdot \Delta x + f(x_2) \cdot \Delta x + f(x_3) \cdot \Delta x + \dots + f(x_n) \cdot \Delta x$$

$$= \sum_{i=1}^n f(x_i) \cdot \Delta x$$

Example: $f(x) = x + 2$ with $P = \{0, 2, 4, 6, 8\}$; $\Delta x = \frac{8-0}{4} = 2$ Example: $f(x) = x + 2$ with $P = \{0, 2, 4, 6, 8\}$; $\Delta x = \frac{8-0}{4} = 2$



$$\begin{aligned} n &= 4 \\ L_n &= f(0) \cdot \Delta x + f(2) \cdot \Delta x \\ &\quad + f(4) \cdot \Delta x + f(6) \cdot \Delta x \\ &= 2 \cdot \Delta x + 4 \cdot \Delta x + 6 \cdot \Delta x + 8 \cdot \Delta x \\ &= (2 + 4 + 6 + 8) \cdot \Delta x \\ &= (2 + 4 + 6 + 8) \cdot 2 = 40 \end{aligned}$$



$$\begin{aligned} n &= 4 \\ R_n &= f(2) \cdot \Delta x + f(4) \cdot \Delta x \\ &\quad + f(6) \cdot \Delta x + f(8) \cdot \Delta x \\ &= 4 \cdot \Delta x + 6 \cdot \Delta x + 8 \cdot \Delta x + 10 \cdot \Delta x \\ &= (4 + 6 + 8 + 10) \cdot \Delta x \\ &= 28 \cdot 2 = 56 \end{aligned}$$

4. Review: Sums and Powers of Integers.

(a) The sum of n integers is given by $\sum_{i=1}^n i = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

(b) The sum of consecutive integers squared is given by $\sum_{i=1}^n i^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

5. Review: Properties of Sigma Notation.

Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ represent two sequences of terms and let c be a constant.

(a) $\sum_{i=1}^n c = nc$

(b) $\sum_{i=1}^n (ca_i) = c \sum_{i=1}^n a_i$
 $= ca_1 + ca_2 + ca_3 + \dots + ca_n$
 $= c(a_1 + a_2 + a_3 + \dots + a_n)$
 $= c \sum_{i=1}^n a_i$

(c) $\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$

(d) $\sum_{i=1}^n (a_i - b_i) = \sum_{i=1}^n a_i - \sum_{i=1}^n b_i$

$\underbrace{c + c + c + \dots + c}_{n \text{ terms}} = nc$

6. Example.

(a) Find the sum of the terms $(i-3)^2$ for $i = 1, 2, \dots, 200$.

$$\begin{aligned} \sum_{i=1}^{200} (i-3)^2 &= \sum_{i=1}^{200} (i^2 - 6i + 9) = \sum_{i=1}^{200} i^2 - \sum_{i=1}^{200} 6i + \sum_{i=1}^{200} 9 \\ (i-3)(i-3) &= i^2 - 6i + 9 \\ &= \frac{200 \cdot (200+1) \cdot (2 \cdot 200 + 1)}{6} - 6 \cdot \frac{200 \cdot (200+1)}{2} + 200 \cdot 9 \\ &= 2,567,900 \end{aligned}$$

(b) Find the sum of the terms $j(j+1)$ for $j = 0, 1, \dots, 9$.

$$\begin{aligned} \sum_{j=0}^9 j(j+1) &= \sum_{j=0}^9 (j^2 + j) = \sum_{j=0}^9 j^2 + \sum_{j=0}^9 j \\ &= \sum_{j=1}^9 j^2 + \sum_{j=1}^9 j \\ &= \frac{9 \cdot (9+1) \cdot (2 \cdot 9 + 1)}{6} + \frac{9 \cdot 10}{2} = 330 \end{aligned}$$